k1 pairing domains: actual torsion versus quotient

Actual curve. E/F7: $y^2=x^3+3x+2$ is nonsingular. Lean identifies the entire rational point group with Z/9, generated by G=(0,3). Rational3torsion A and actual B=E(F7)/3E(F7) both have order3, but the natural A→B map is zero and G has nonzero class. Embeddingdegree1;3 divides7−1 and3^2 divides#E9. Actual geometric3torsion outside the rational image is proved. Signed standing: unverified. [pcc:unverified]

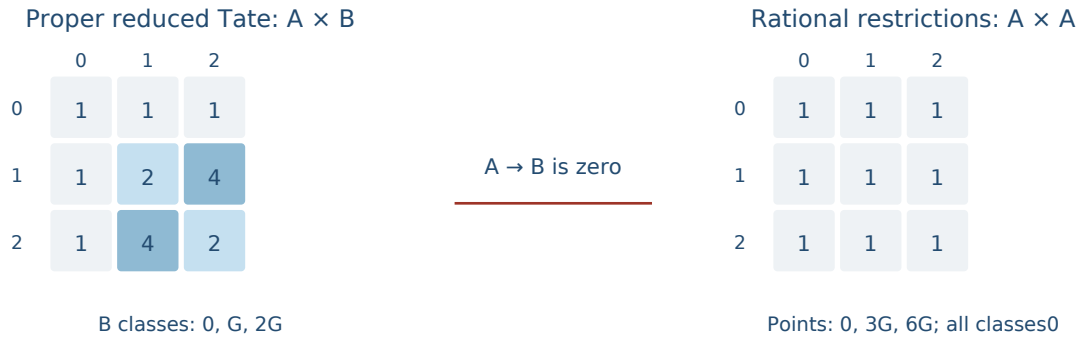

Figure. Rows are U=u·3G, u=0,1,2. The proper map gives T(3G,G)=2, a primitive cube root. Both rational-torsion restrictions give1: the Weil expression is trivial on A×A, and every second-slot torsion point represents class0 for Tate. These are corollaries of checked actual Miller values, not a native experiment. [pcc:unverified]

Proper objects and target. Raw actual tangent/Miller ratios descend to the true $F7^*/(F7^*)^3$ quotient. Finalexponent2 has kernel precisely the cubes and gives an isomorphism to the actual unit cube roots; both targets have order3. Raw/reduced maps on A×B are bilinear and nondegenerate in both slots. Every actual point representative and every valid auxiliary gives the same targetclass; no torsion-as-quotient-representative assumption is used. [pcc:unverified]

Skeptical checks. Actual point coverage is exhaustive. All27Tate point pairs and243candidate auxiliary triples, and9Weil torsion pairs and81candidate auxiliary triples, are kernel checked with the valid-support conditions. Every used numerator/denominator is nonzero. Two direct negations refute the unrestricted rational restriction readings. More generally, every alternating biadditive map on this actual cyclic rational subgroup is trivial, and any normalized quotient pairing pulled back along A→B loses detection. [pcc:unverified]

Geometry boundary. Exact triple-intersection tangent polynomials, the actual line-evaluation match and algebraic infinity-chart identities are proved. Canonical divisor/valuation pairing identification and generic MOV/Frey–Rück theorems remain open. These finite expressions confirm the qualified quotient-domain behavior; no qualified source theorem is refuted. [pcc:unverified]

Checks. Whole-project gate 116/0failures/0trusted; Lean4.35.0/Mathlib56a00585…. Only standard foundations; no holes, custom axioms, native evaluator, raised limits, global elaboration flags or waivers. Complete main binding/source graph, failed attempts and gate records are pinned in sources.json/proof-map.json/evidence. No native experiment, paid compute or agent upload. [pcc:unverified]

Sources. Galbraith chapter26,26.1/26.2, support/quotient definitions and26.3.3; PDFpage577 visually inspected. Existing actual F7/F25 and pinned Mathlib group/quotient/unit APIs reviewed. Original FR1994 remains unread after bounded alternate DOI/catalog/JSTOR routes; JSTOR200was an HTML challenge. KB unavailable, bounded fallback/no novelty claim. Other fourteen-item obligations remain explicit. [pcc:unverified]